LECTURE 01

Introduction to Theory of Computation

Course Code : CSE 3103
Course Title : Theory of Computation

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INTRODUCTION

THEORY OF COMPUTATION

The theory of computation is the branch that deals with **what problems can be solved** on **a model of computation**, using an algorithm, **how efficiently they can be solved** or to what degree.

3 CENTRAL AREA.

1. AUTOMATA THEORY
2. COMPUTABILITY THEORY
3. COMPLEXITY THEORY

They are linked by the question:

- What are the fundamental **capabilities and limitations** of computers?

AUTOMATA THEORY

AUTOMATA THEORY

Dealing with developing formal **mathematical models of computation** that reflect real-world **computers.** Other words, representing a physical machine on paper.

COMPUTER/AUTOMATON

Computing Device or Machine. For example: Laptop, Desktop, Calculator etc.

MATHEMATICAL MODELS OF COMPUTATION

- Finite Automata
- Push Down Automata
- Context free grammar
- Turing Machine

COMPUTABILITY THEORY

COMPUTABILITY THEORY

The classification of problems is by those that are solvable and those that are not.

Problems

Computable by Designed Automata	Not Computable by Designed Automata

Certain basic problems cannot be solved by computers. One example of this phenomenon is the problem of determining **whether a mathematical statement is true or false.** No computer algorithm can perform this task.

COMPLEXITY THEORY

<u>COMPLEXITY THEORY</u>

The objective is to classify problems as easy ones and hard ones.

Computable Problems

Computationally Easy problem	Computationally Hard problem

Easy Problem: Sort a list of numbers in ascending order.

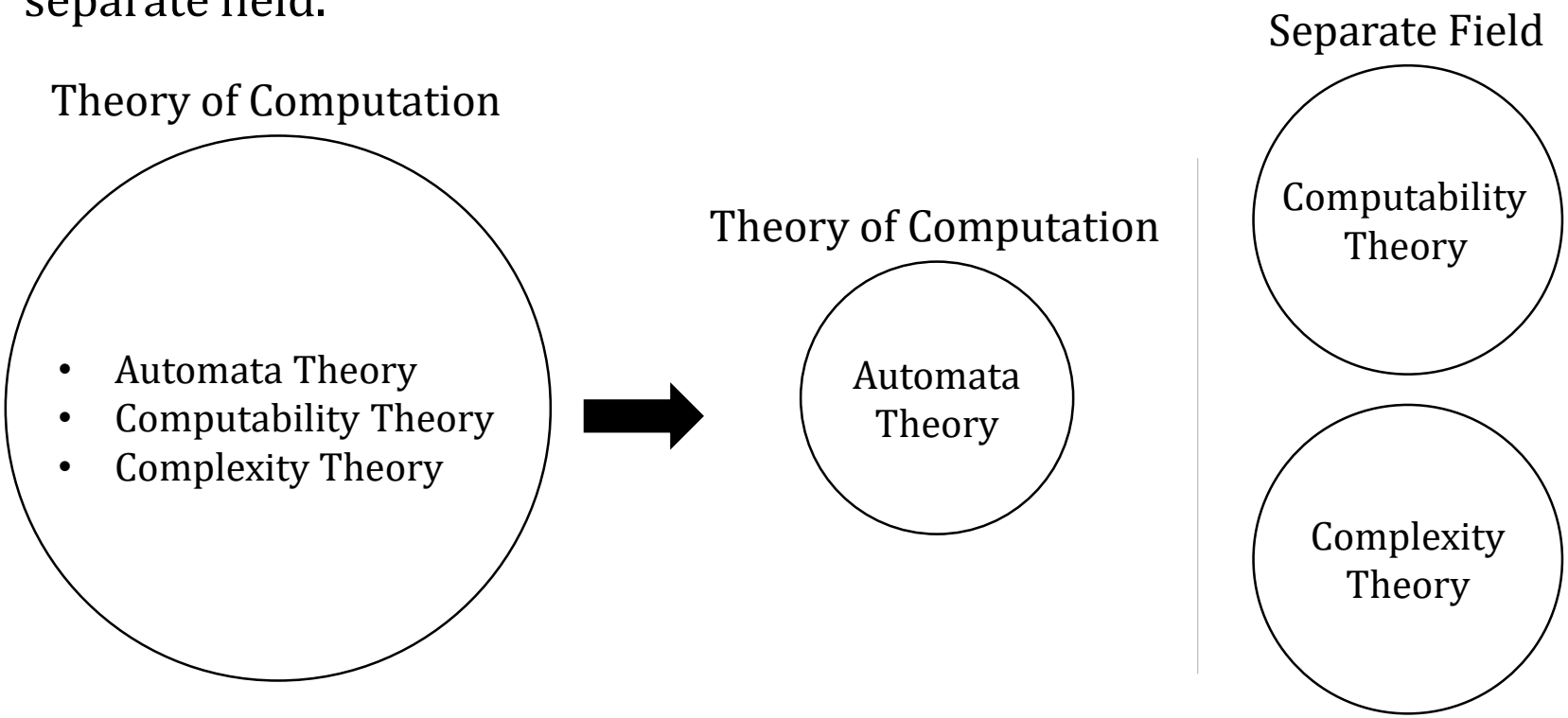
Hard Problem: Schedule of classes for the entire university to satisfy some reasonable constraints.

- **What makes some problems computationally hard and others easy?**

This is the central question of complexity theory.

THEORY OF COMPUTATION

In Theory of Computation, we will mostly discuss about Automata Theory. Computability and Complexity theory are discussed in the separate field.



Hence, Theory of computation is also called Automata Theory.

IN SUMMARY

Theory of computation or Automata theory

Dealing with developing formal mathematical models of computation that reflect real-world computers. Other words, representing a physical machine on paper.

We will design

computer or computing machine

through out this course.

IMPORTANCE OF THIS COURSE

There are several reasons why the study **Theory of Computation.**

- Finite Automata are a useful model for many important kinds of **hardware and software.**
- Students are expected to learn about software for designing and checking the behavior of **digital circuits.**
- They can explain:
  - **lexical analyzer,**
  - **syntax analyzer and**
  - **semantic analyzer**

  of a typical compiler.